Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal

B. 40 Kcal, 5 Kcal

C. 30 Kcal, 8.58 Kcal

D. 25 Kcal, 8 Kcal

E. 29 Kcal, 7.77 Kcal

F. 34 Kcal, 6.9 Kcal

G. 30 Kcal, 6.45 Kcal

H. 32 Kcal, 7 Kcal

I. 35 Kcal, 7.5 Kcal

J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

to calculate the quantity of heat required, we need to use the formula q = mcÎĶT, where q is the heat required, m is the mass of the substance, c is the specific heat capacity, and ÎĶT is the change in temperature.

for (a), the mass of 1 liter of water is approximately 1 kg, so m = 1000 g, c = 1 cal/gÂ°C, ÎĶT = 60 Â°C - 30 Â°C = 30 Â°C.

the heat required to heat the water is q = mcÎĶT = 1000 g * 1 cal/gÂ°C * 30 Â°C = 30000 cal = 30 Kcal.

for (b), the mass of the block of aluminum is 1 kg = 1000 g, c = 0.215 cal/gÂ°C, ÎĶT = 60 Â°C - 30 Â°C = 30 Â°C.

the heat required to heat the aluminum is q = mcÎĶT = 1000 g * 0.215 cal/gÂ°C * 30 Â°C = 6425 cal = 6.425 Kcal.

</think><answer> C </answer>